

CPG Sales Promotion Bot



INTRODUCTION

SMARTER PROMOS. STRONGER GROWTH.

A five-agent bot that turns raw sales data into a decision-ready insight.

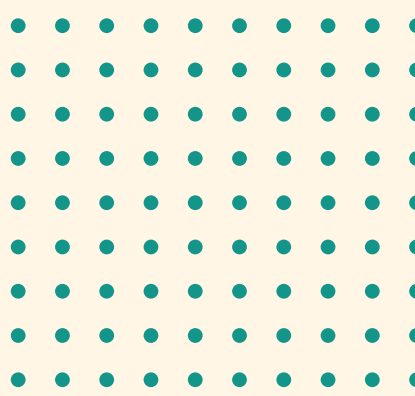
• UiPath Orchestration

• Anthropic Claude

• Google Sheets

PROBLEM

Category managers drown in **spreadsheets and hunches.**



"Every promo review is a full day of pivot tables, and the insight lands after the window has already closed."

CPG CATEGORY MANAGER

1

5-6 hours

Manual effort to review one promotion cycle

2

15–20%

Uplift lost to guessed discount & timing choices

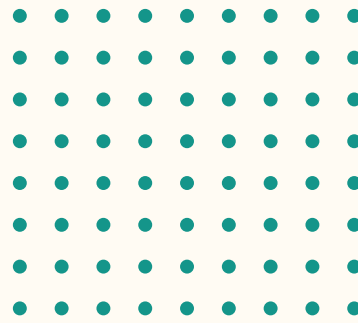
3

Minimum

Audit trail. Limited record of how a number was reached

GOAL

Turn raw sales data into a **decision-ready insight**, automatically.



The bot answers four questions a category manager actually asks, then hands back a written recommendation, all in a single run.



MEASURE

How much did promotions lift sales?

Compute promo vs. no-promo uplift overall, and break it down by category, channel, and country.



DETECT

What kind of promotion works best?

Compare deep BOGO-style discounts against flat ones, and test weekend and holiday timing effects.



REASON

What should we do next?

Hand the numbers to Claude for a plain-English executive summary plus two concrete recommendations.

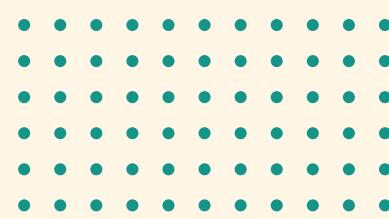


DELIVER

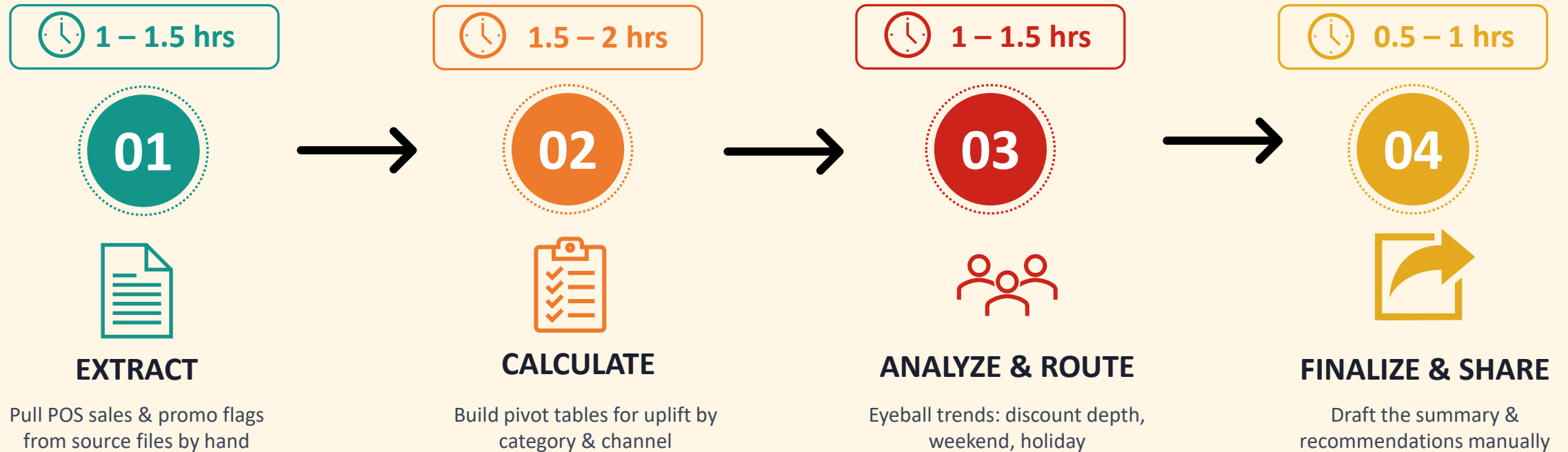
Where does it land?

Append a timestamped result row to a shared Google Sheet.
A living dashboard, not a one-off report.

MANUAL PROCESS



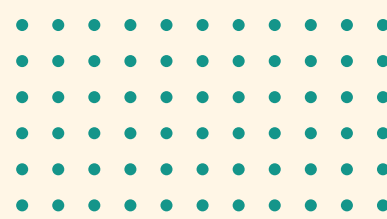
How a promo gets reviewed **today**.



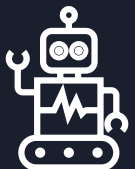
Total Time: 5-6 hours per promotion

Long cycle time - Heavy manual effort – High error risk – Lack of visibility – Delayed go-to-market – Frustrated teams and partner

AUTOMATED PROCESS

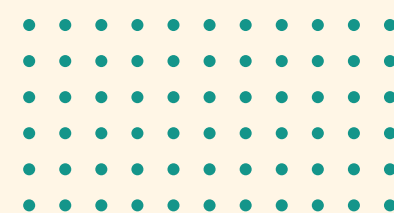


The bot does the same work in **one orchestrated run**.



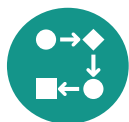
A linear deterministic sequence (Start → Agent 1 → Agent 2 → Agent 3 → Agent 4 → Agent 5 → End), so every run is identical and fully traceable.

SOLUTION ARCHITECTURE



Four building blocks, **one connected pipeline.**

UiPath Studio orchestrates the run as an Agentic Process (BPMN). Two connectors call external services; two script tasks do the maths in-process.



ORCHESTRATION

UiPath Studio

Process-orchestration BPMN with five named tasks in a deterministic Start→End sequence.



DATA IN

Drive + HTTP

An HTTP GET pulls the sales CSV straight from a Google Drive download link.



REASONING

Anthropic Claude

The claude-sonnet-4.6 messages API turns the metric bundle into a written insight.



DATA OUT

Google Sheets

A WriteRow call appends KPIs + the AI summary to a shared dashboard tab.

Two script tasks (Agents 2 & 3) run JavaScript inside the workflow, so there is no external compute, no extra latency.

INPUT

One CSV of **sales transactions**.



DATASET

CPG Multi-Country Sales Dataset

Public dataset on Kaggle, hosted as a CSV on Google Drive for the run.

[Kaggle](#) ↗

How the data is read

- ✓ promo_flag = 1 marks a promoted transaction
- ✓ Uplift = (avg promo units – avg non-promo units) ÷ avg non-promo, as a %
- ✓ discount_pct >= 0.25 is treated as a deep "BOGO-style" deal; below that is "flat"
- ✓ Grouped three ways: by category, channel, and country
- ✓ Timing splits: weekend vs weekday and holiday vs non-holiday

store_id	channel	sku_id	sku_name	category	subcategory	brand
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB
STORE0001	Hypermarket	SKU0086	BrandB Shampoo	Personal Care	Shampoo	BrandB

Columns the agents actually read

```
units_sold · promo_flag · discount_pct
category · channel · country
is_weekend · is_holiday
```


AGENT DESIGN

Five agents, **each doing one job well.**

The orchestration runs them in a fixed line. Each section below breaks one agent down: task type, inputs, logic, output, and guardrails.



Agent 1

Load CSV Data



Agent 2

Promotion Analysis



Agent 3

Trend Detection



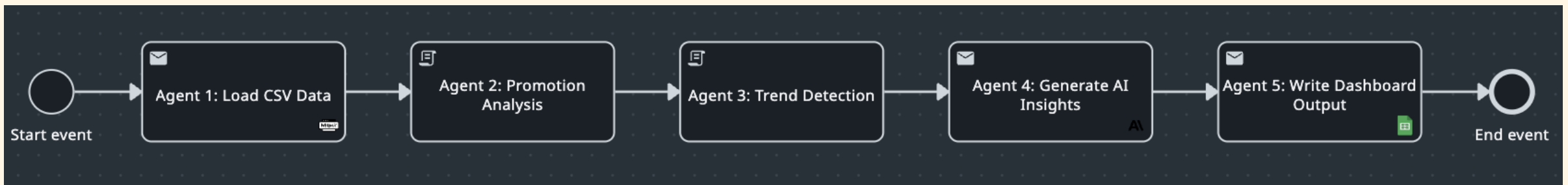
Agent 4

**Generate AI
Insights**



Agent 5

**Write Dashboard
Output**

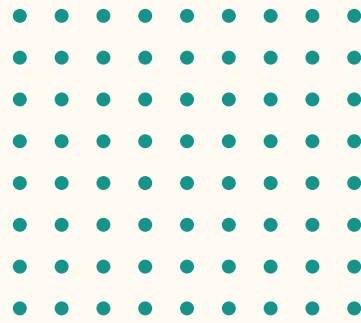




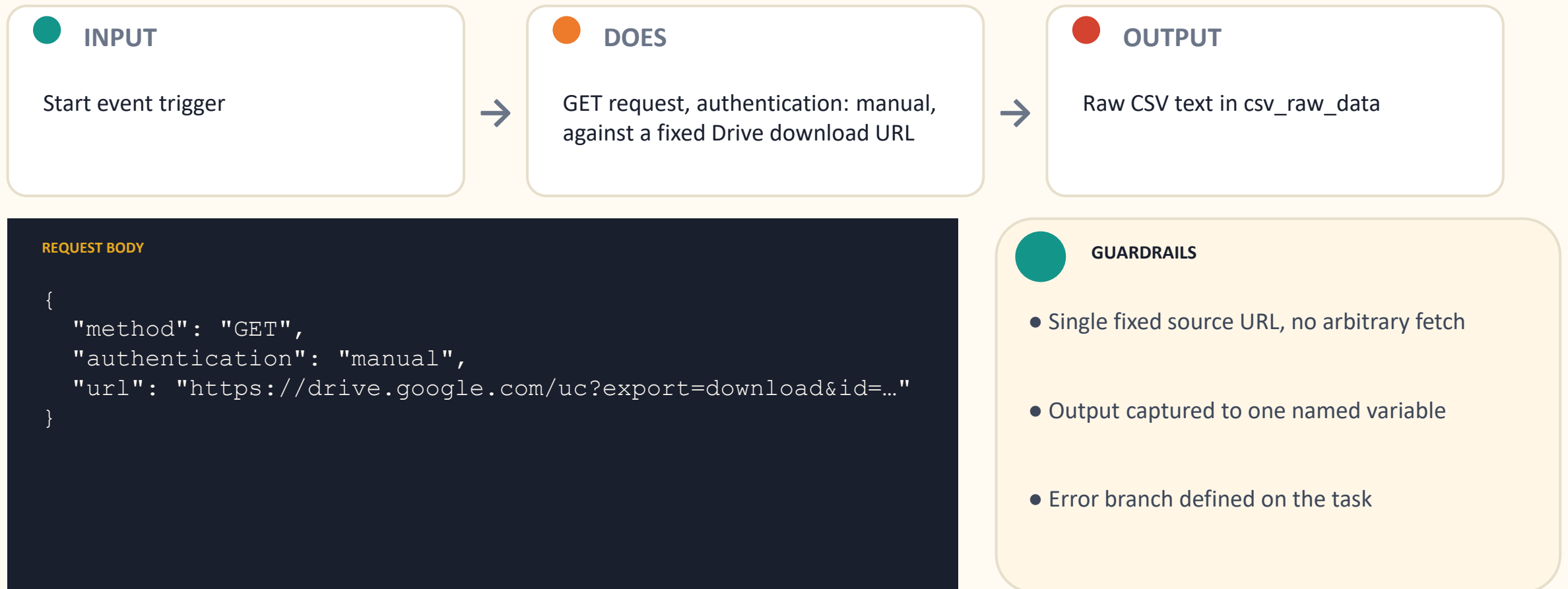
AGENT 01

Load CSV Data

Send Task · HTTP connector



The pipeline's entry point: one HTTP GET pulls the CSV from a Google Drive link and stores it as raw text for every downstream agent.



Promotion Analysis

Script Task · in-process JavaScript

The quantitative core: parses the CSV, splits promoted from non-promoted units, and computes percentage uplift overall, per category and channel.



INPUT

promo_flag units_sold category
channel country



DOES

Averages promo vs. non-promo units,
then takes the percentage gain,
grouped four ways



OUTPUT

scriptResponse: overall + per-group
uplift

CORE LOGIC

```
// split units by promo flag
if (isPromo) promoUnits.push(u);
else        noPromoUnits.push(u);

// uplift = % gain of promo avg
uplift = (avg(promo) - avg(noPromo))
         / avg(noPromo) * 100;
```



GUARDRAILS

- Empty groups return "0", never divide by zero
- Runs in-process, no external call to fail
- Pure function of the CSV, fully reproducible

3

AGENT 03

Trend Detection

Script Task · in-process JavaScript

Looks only at promoted rows and asks which mechanics and timing move volume, comparing average units across three binary splits.



INPUT

discount_pct is_weekend is_holiday
(promo rows only)



DOES

Buckets units by deep vs. flat (0.25
cutoff), weekend vs. weekday, holiday
vs. not



OUTPUT

scriptResponse2: avg units per split

CORE LOGIC

```
// promo rows only
if (promo_flag !== 1) continue;

disc >= 0.25 ? bogo.push(u)
              : flat.push(u);
isWeekend ? weekend.push(u)
           : weekday.push(u);
isHoliday ? holiday.push(u)
           : nonHoliday.push(u);
```



GUARDRAILS

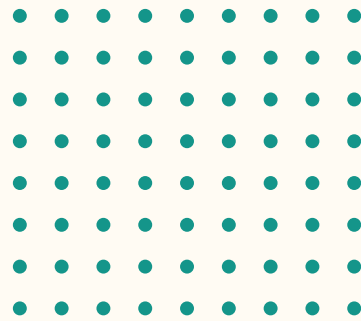
- Skips non-promo rows explicitly
- Averages guard against empty buckets
- Threshold isolated as one constant (0.25)



AGENT 04

Generate AI Insights

Send Task · Anthropic Claude (messages)



The reasoning layer. Feeds computed figures to Claude and asks for a tight executive summary plus two actionable recommendations.



INPUT

overall_uplift, flat_avg_units,
bogo_avg_units, category breakdown



DOES

Calls claude-sonnet-4-6 with a fixed
analyst prompt (temp 1, 1024 tokens)



OUTPUT

Summary + 2 recommendations as text

THE PROMPT

```
"You are an CPG analyst. Overall promo uplift is " +  
vars.scriptResponse.overall_uplift + "%". Flat discount  
averages " + vars.scriptResponse2.flat_avg_units + " units  
vs BOGO at " + vars.scriptResponse2.bogo_avg_units + "  
units. Category breakdown: " +  
vars.scriptResponse.category_uplift + ".
```

```
Generate: 1) 2-sentence executive summary. 2) Two actionable  
recommendations."
```



GUARDRAILS

- Templated prompt, only numbers vary per run
- Token cap (1024) bounds output length and cost
- Claude sees only aggregates, never raw records

5

AGENT 05

Write Dashboard Output

Send Task · Google Sheets (WriteRow)

The delivery step: appends one timestamped row to a shared Google Sheet. In APPEND mode the sheet doubles as the audit log.



INPUT

Date, overall uplift, per-category uplift values



DOES

WriteRow in APPEND mode to Sheet1!A1:Z1000, headers off



OUTPUT

New row on the shared dashboard sheet

ROW WRITTEN

```
[ new Date().toISOString().split('T')[0],  
  overall_uplift,  
  cat_personal_care,  
  cat_dairy,  
  cat_home_care,  
  ... ]
```



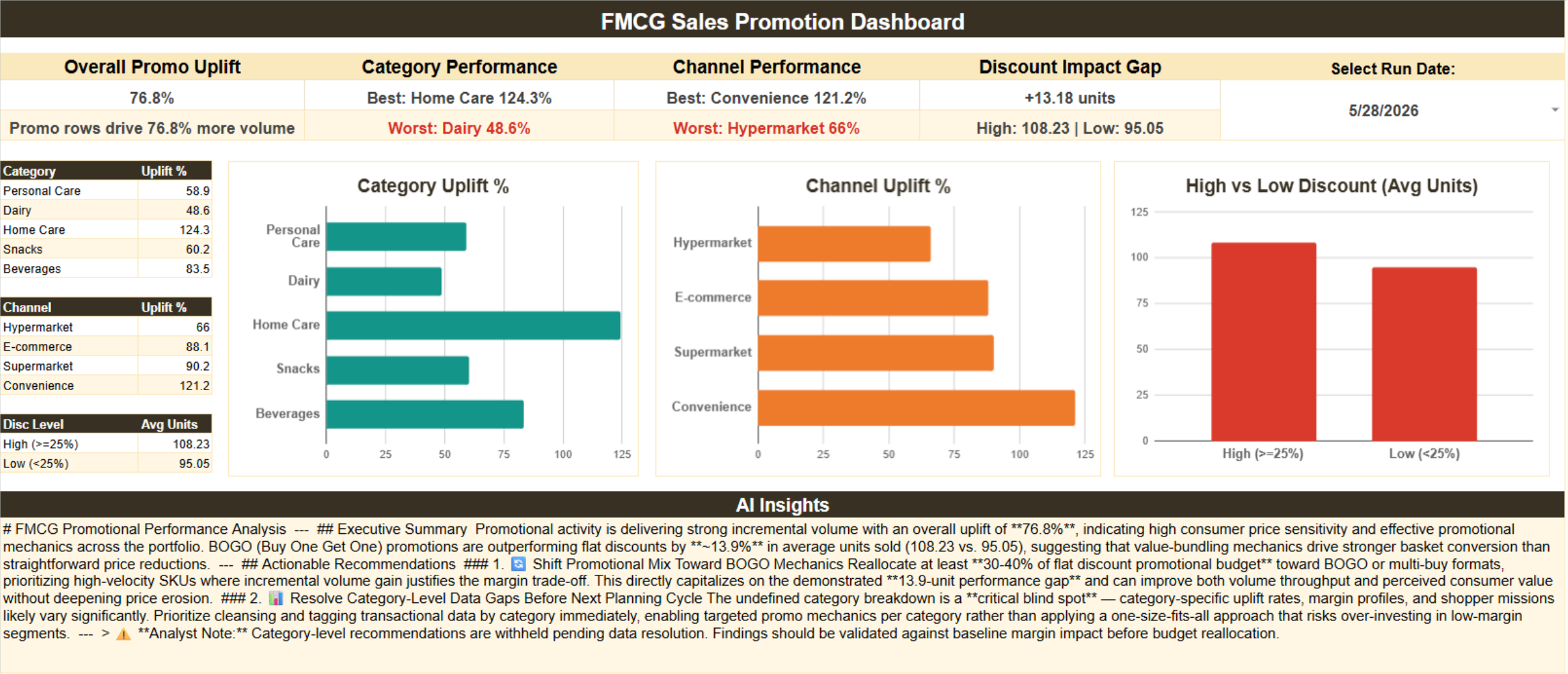
GUARDRAILS

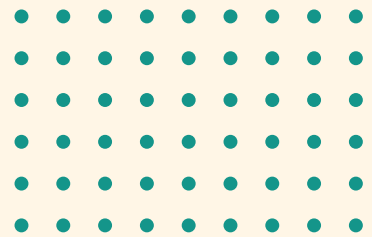
- APPEND only, never overwrites history
- Each row carries its own date stamp
- Bounded range keeps writes predictable

DEMO



OUTPUT





ADVANTAGES

Why the agentic split **works**.



SPEED

A full day of analysis collapses into **one automated run**



REPRODUCIBLE

Deterministic BPMN sequence, the **same inputs give the same output**



SEPARATION OF CONCERNS

Maths lives in **script tasks**; judgment lives in **Claude**



LIVING DASHBOARD

Append-only Sheets log builds a **history of every run**

LIMITATIONS

A working POC. **Honest about what it isn't yet.**

01

Hard-coded categories

Agent names each category explicitly;
a new category needs a code change.

02

Limited data volume

Processes a modest CSV in memory;
large datasets would need batching or
a DB backend.

03

Uplift $\neq$ causation

Promo vs. non-promo averaging
ignores seasonality and baseline drift.

04

Single CSV source

Read one fixed Drive link, with no live
data update yet

05

Desktop runtime

Runs from Studio; no Orchestrator
schedule or Robot fleet.

06

Default model settings

Temperature 1, 1024 max tokens, not
yet tuned for terse, stable output.



Five agents. One run. A
decision-ready insight.

Thank you.



LINKS

- **Original Dataset Kaggle:** <https://www.kaggle.com/datasets/robertocarlost/fmcg-multi-country-sales-dataset?>
- **GitHub:** https://github.com/moq32/CPG_Sales_Promo_Bot/tree/master
- **Dashboard Link:** <https://docs.google.com/spreadsheets/d/1cP5aEyrdMUFWgMTgSV7hoNi6dqQFPEwmhGADiNtyGIA/edit?gid=776298885#gid=776298885>